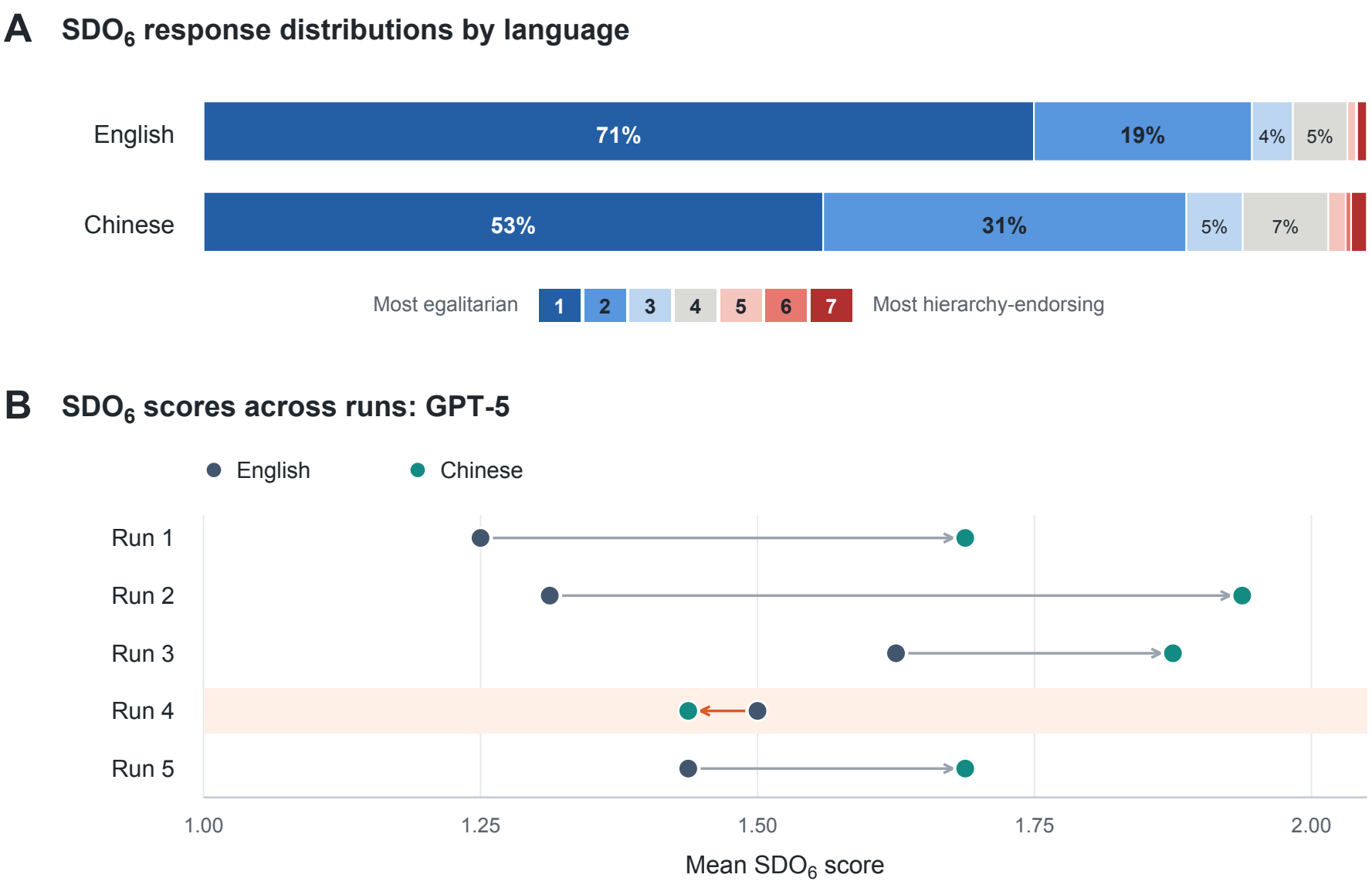


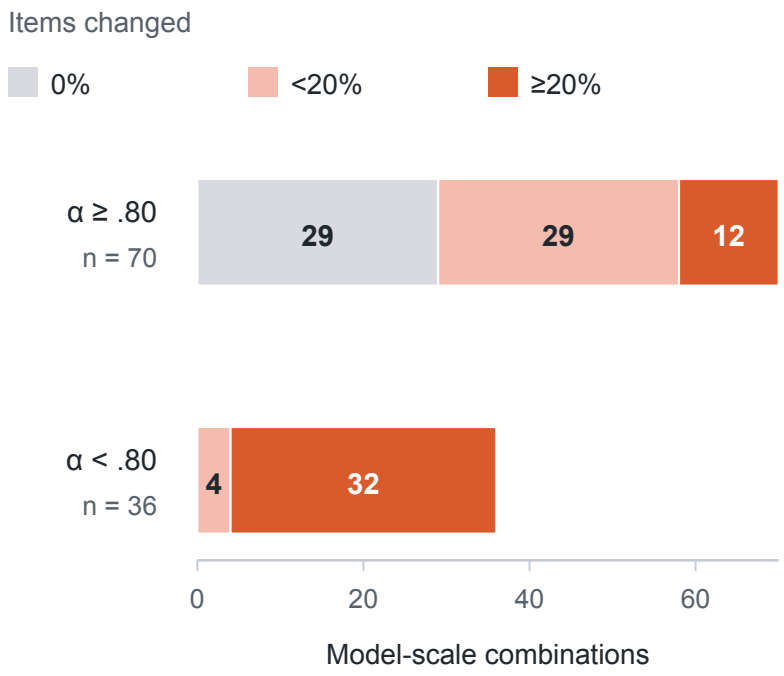
Fig. 2. Models gave less strongly egalitarian responses in Chinese on average, but the direction of the language difference varied across runs for some models. (A) Proportion of SDO_6 responses at each scale point, shown separately by language and pooled across 27 models, 16 items and five runs. Reverse-keyed items were rescored so that 1 indicates the most egalitarian response and 7 the most hierarchy-endorsing response. Percentages use all valid responses within each language (English, n = 2,156; Chinese, n = 2,146). **(B)** GPT-5's mean SDO_6 score in English and Chinese in each of five runs. In each run, GPT-5 answered all 16 items once in each language; the score is the mean of the 16 rescored responses. Arrows connect the English score to the Chinese score. Orange highlights Run 4, in which the direction of the language difference reversed. Model-level patterns across runs are shown in Supplementary Fig. S14.

A VSA responses across runs: GPT-5.5

	Run 1	Run 2	Run 3	Run 4	Run 5	Changed
Item 1	-1	-1	-1	**0**	-1	**Yes**
Item 2	-3	-3	-3	-3	-3	No
Item 3	0	0	0	0	0	No
Item 4	**-2**	-3	-3	-3	-3	**Yes**
Item 5	**-1**	0	0	0	0	**Yes**
Item 6	0	0	0	0	0	No

Krippendorff's α = 0.94 **3 of 6 items changed**

B Reliability and response changes



C SDO response variation by initial response

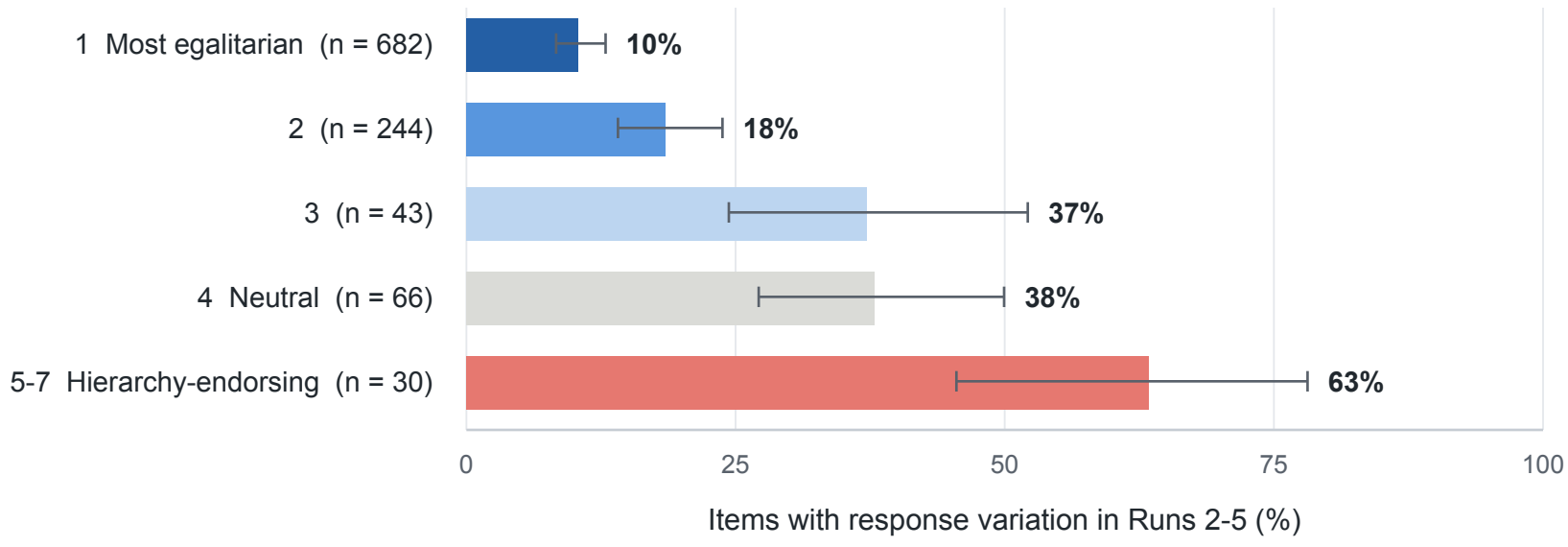
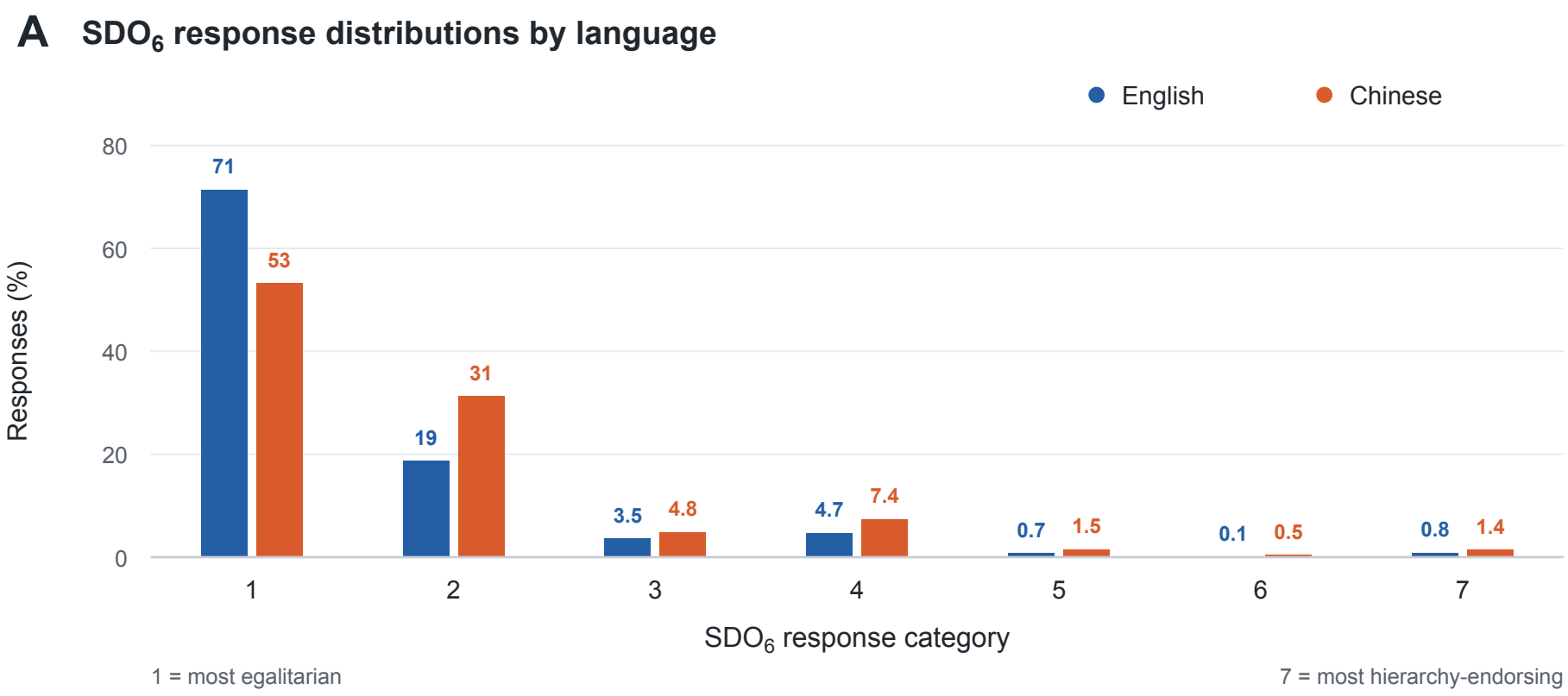


Fig. 3. High reliability did not imply identical responses across repeated runs. (A) GPT-5.5's rescored responses to the six Very Short Authoritarianism (VSA) items across five runs. In each run, the model answered each item once on the -4 to +4 response scale. Orange cells mark responses that differ from the modal response for that item; shaded rows identify items with at least one response change. Krippendorff's interval α was .94, exceeding the .80 benchmark, although three of six items changed across runs. **(B)** Model-scale combinations grouped by whether Krippendorff's interval α met the .80 benchmark and by the proportion of items with at least one response change across five runs. The four scale versions were SDO_7(s) in English, SDO_6 in English, SDO_6 in Chinese and VSA in English. Of 108 combinations (27 models × four versions), two had invariant responses and undefined α and were omitted. Among the 70 combinations with α ≥ .80, 41 changed at least one item, including 12 that changed at least 20% of items. **(C)** Proportion of model-item combinations with at least two distinct responses among Runs 2-5, grouped by the rescored response in Run 1. All 27 models and the three social dominance orientation (SDO) scale versions were pooled; only model-item combinations with five valid responses were included. Run 1 defines the groups and is excluded from the response-variation calculation. Values of n indicate model-item combinations per group; error bars show 95% Wilson intervals for the pooled proportions.



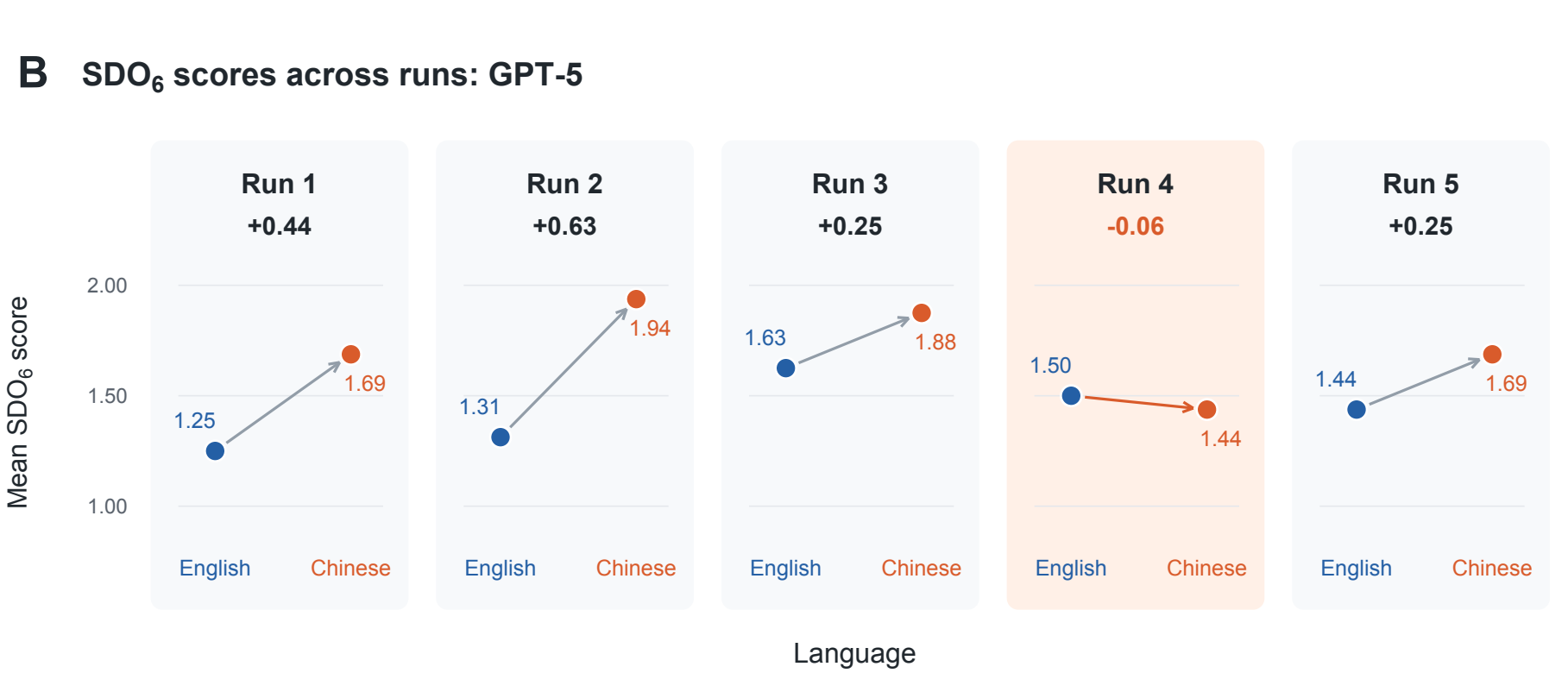


Fig. 2. Models gave less strongly egalitarian responses in Chinese on average, but the direction of the language difference varied across runs for some models. (A) Proportion of SDO_6 responses at each scale point, shown separately by language (English, blue; Chinese, orange) and pooled across 27 models, 16 items and five runs. Reverse-keyed items were rescored so that 1 indicates the most egalitarian response and 7 the most hierarchy-endorsing response. Percentages use all valid responses within each language (English, n = 2,156; Chinese, n = 2,146). **(B)** GPT-5's mean SDO_6 score in English and Chinese in each of five runs. In each run, GPT-5 answered all 16 items once in each language; the score is the mean of the 16 rescored responses. In each run-specific plot, the horizontal axis identifies language and the vertical axis shows the mean SDO_6 score. In each small multiple, an arrow connects the English score to the Chinese score; signed values indicate the Chinese score minus the English score, calculated before rounding. Orange highlights Run 4, in which the direction of the language difference reversed. Model-level patterns across runs are shown in Supplementary Fig. S14.

Alternative charts | Figure 3

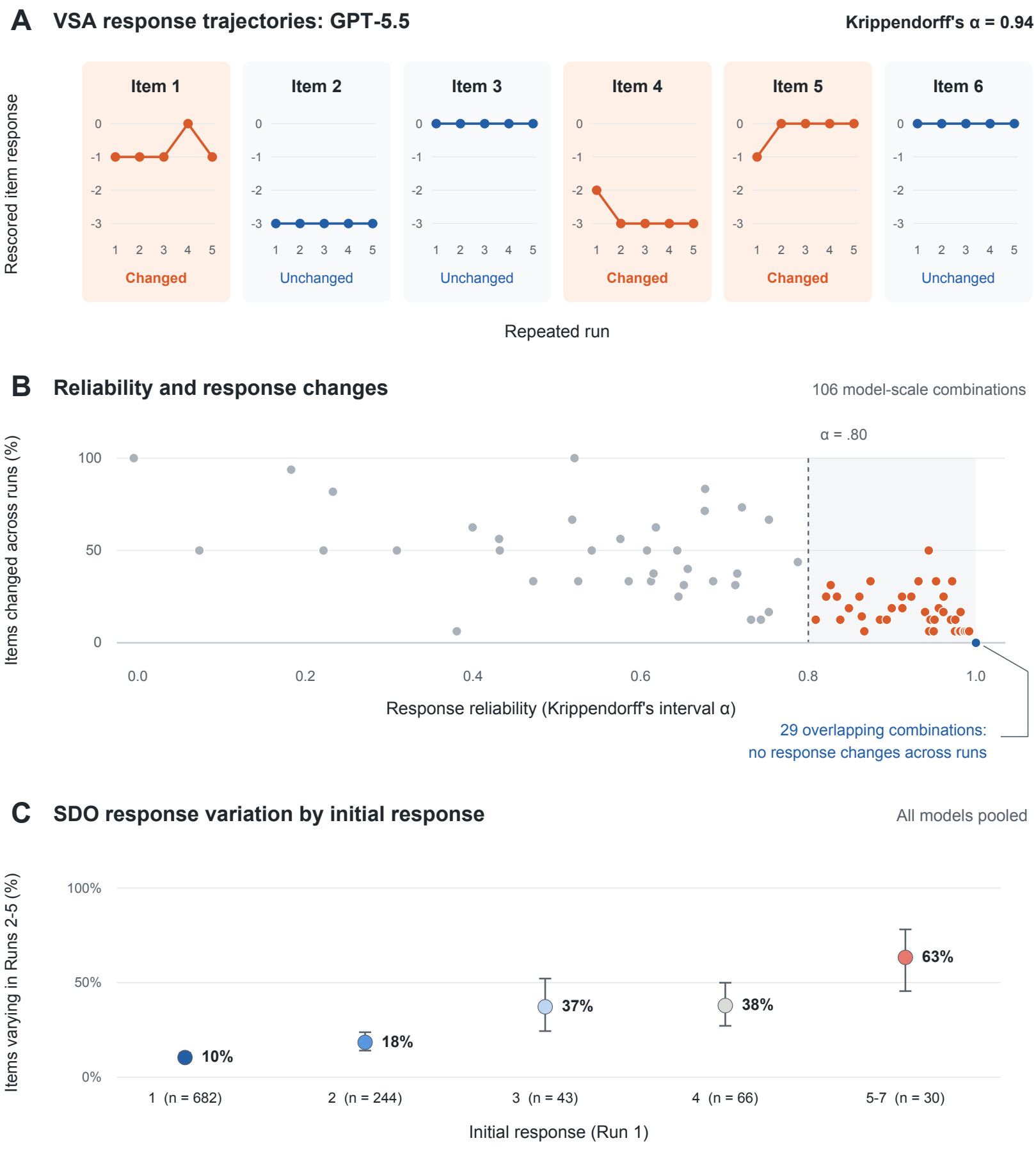


Fig. 3. High reliability did not imply identical responses across repeated runs. (A) GPT-5.5's rescored responses to the six Very Short Authoritarianism (VSA) items across five runs. In each run, the model answered each item once on the -4 to +4 response scale. Each small multiple shows one item's response trajectory: the horizontal axis is repeated run (1-5), and the vertical axis is the rescored response to that individual item, rather than an average. All items use a shared vertical scale; orange indicates items with at least one response change, and blue indicates unchanged items. Krippendorff's interval α was .94, exceeding the .80 benchmark, although three of six items changed across runs. **(B)** Points show model-scale combinations, positioned by Krippendorff's interval α and the percentage of items with at least one response change across five runs. The dashed line marks α = .80. Orange points meet this benchmark despite response changes; blue points meet it without changes; grey points fall below it. All markers have the same size. The labelled point at α = 1.0 and 0% response change represents 29 combinations at identical coordinates; the label identifies their overlap. The four scale versions were SDO_{7}(s) in English, SDO_{6} in English, SDO_{6} in Chinese and VSA in English. Of 108 combinations (27 models × four versions), two had invariant responses and undefined α and were omitted. Among the 70 combinations with α ≥ .80, 41 changed at least one item, including 12 that changed at least 20% of items. **(C)** Proportion of model-item combinations with at least two distinct responses among Runs 2-5, grouped by the rescored response in Run 1. All 27 models and the three social dominance orientation (SDO) scale versions were pooled; only model-item combinations with five valid responses were included. Run 1 defines the groups and is excluded from the response-variation calculation. Values of n indicate model-item combinations per group; error bars show 95% Wilson intervals for the pooled proportions.

Supplementary figure | Former Figure 2C



Supplementary Fig. S14. Language-effect direction across repeated runs. Each tile represents one of the 27 models, grouped by model family and coloured according to the direction of its mean Chinese-minus-English SDO_6 score difference across five runs. Scores use rescored valid responses. Twenty models scored higher in Chinese in every run; one scored at least as high in Chinese in all runs, with one tie; one scored at least as high in English in all runs, with one tie. Orange identifies the five models with both positive and negative differences across runs. This panel was moved from the original Figure 2C; S14 continues the existing S1-S13 supplementary numbering.